# Christina Shin

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**INTERESTS**

Computer Networking, Mobile Computing, AR/VR Systems, Volumetric Video Streaming, Autonomous & Connected Vehicles, 3D Data Processing, 3D Mapping

**EDUCATION**

**University of Southern California**, Los Angeles, California

- **Ph.D. Candidate** in Computer Science — Jan 2026 (Expected)
  *Dissertation (in progress): Harnessing Vehicular 3D Primitives for Networked Scene Operation and Manipulation*

**Ewha Womans University**, Seoul, Republic of Korea

- **M.S.** in Computer Science and Engineering — Feb 2019
  *Thesis: Network Diagnosis and Reconstruction in Vehicular Ad-Hoc Networks*
- **B.S.** in Computer Science and Engineering — Feb 2017

**EXPERIENCE**

**General Motors**, Warren, Michigan

Research Intern & Collaborator (Mentors: Fan Bai and Chuan Li) — May 2021 – Present

- Led the design and implementation of a volumetric video delivery system for vehicles, enabling low-latency and immersive AR passenger experiences.
- Developed a 3D traffic reconstruction system using multi-vehicle point cloud registration, generating volumetric videos of dynamic traffic scenes to improve scalability and scene understanding; contributed to a U.S. patent application (US20230140324A1, now abandoned).

**Networked Systems Lab**, University of Southern California

Research Assistant (Advisor: Prof. Ramesh Govindan) — Aug 2019 – Present

- Invented an infrastructure-assisted autonomous driving system that leverages roadside LiDARs and edge compute to extend vehicle perception beyond occlusions.
- Devised a 3D building reconstruction system with a LiDAR-equipped UAV, optimizing path planning for data capture and generating near real-time SLAM-based 3D models of buildings.
- Contributed to a live volumetric video streaming system, focusing on the 3D video rendering pipeline to ensure real-time visualization of captured scenes.

**Intelligent Networked Systems Lab**, Ewha Womans University

Research Assistant (Advisor: Prof. HyungJune Lee) — Jan 2017 – May 2019

- Designed a traffic density estimation algorithm using opportunistic V2V packet probing under strict time deadlines.
- Proposed a multi-UAV relay algorithm for network recovery, improving resilience of ad-hoc networks in failure scenarios.

**PUBLICATION**

**CONFERENCE**

- RECAP: 3D Traffic Reconstruction
  **Christina Suyong Shin**, Weiwu Pang, Chuan Li, Fan Bai, Fawad Ahmad, Jeongyeup Paek, and Ramesh Govindan
  *Proceedings of the ACM on Mobile Computing and Networking (MobiCom)*, 2024.

- Cooperative Infrastructure Perception
  **Christina Suyong Shin***, Fawad Ahmad*, Weiwu Pang*, Branden Leong, Pradipta Ghosh, and Ramesh Govindan (* equal contributions)
  *IEEE/ACM Conference on Internet-of-Things Design and Implementation (IoTDI)*, 2024.
- AeroTraj: Trajectory Planning for Fast and Accurate 3D Reconstruction Using a Drone-based LiDAR
  Fawad Ahmad, **Christina Suyong Shin**, Rajrup Ghosh, John D'Ambrosio, Eugene Chai, Karthikeyan Sundaresan, and Ramesh Govindan
  *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (Ubicomp/IMWUT)*, 2023.
- Progressive ad-hoc route reconstruction using distributed UAV relays after a large-scale failure
  **Christina Suyong Shin**, So-Yeon Park, JinYi Yoon, and HyungJune Lee
  *IEEE Wireless Communications and Networking Conference (WCNC)*, 2018.
- DroneNet+: Adaptive Route Recovery Using Path Stitching of UAVs in Ad-Hoc Networks
  So-Yeon Park, Dahee Jeong, **Christina Suyong Shin**, and HyungJune Lee
  *IEEE Global Communications Conference (GLOBECOM)*, 2017.

### JOURNAL

- Infrastructure-less Vehicle Traffic Density Estimation via Distributed Packet Probing in V2V Network
  **Christina Suyong Shin**, JiHo Lee, and HyungJune Lee
  *IEEE Transactions on Vehicular Technology (TVT)*, vol. 69, no. 10, Oct 2020.
- DroneNetX: Network Reconstruction through Connectivity Probing and Relay Deployment by Multiple UAVs in Ad-Hoc Networks
  So-Yeon Park, **Christina Suyong Shin**, Dahee Jeong, and HyungJune Lee
  *IEEE Transactions on Vehicular Technology (TVT)*, vol. 67, no. 11, Nov 2018.

### UNDER SUBMISSION

- Vehicular AR Streaming. *Submitted to MobiCom 2025.*
- Vehicular AR Tracking. *Submitted to MobiCom 2025.*
- Live Volumetric Video Streaming. *Submitted to CoNext 2025.*

## AWARDS & HONORS

- Annenberg Fellowship, University of Southern California — 2019
  Awarded to outstanding Ph.D. students joining in Fall 2019.
- Qualcomm Innovation Awards, Qualcomm x Ewha — 2017
  Recognized for proposing a lightweight network hole replacement algorithm through UAV-net.
- Silver Prize in Graduation Capstone Design, Ewha Womans University — 2016
  For an outstanding project developing *SimMusic*, a language that encodes music notes and plays them on *Lego Mindstorms NXT*.
- Dean's List, Ewha Womans University — 2013, 2015, 2016
  For attaining a GPA above 3.75/4.3.
- NSF/SIGMOBILE Travel Grants: ACM MobiCom '25, ACM HotMobile '24.

## TEACHING & MENTORING

### TEACHING ASSISTANT

University of Southern California

- Internetworking (CSCI 353) — Fall 2025
- Computer Networks (CSCI 551/651) — Fall 2023

Ewha Womans University

- Computer Architecture (20493-02) Fall 2018
- Arduino Programming (11208-01) Spring 2018
- C Programming (38407-05) Fall 2017
- Programming Language Theory (20499-01, 20499-02) Spring 2017

**MENTORING**

- Alejandra Cortes (High school): 3D perception with point clouds Summer 2023
- Alexander Due (Undergrad): Object-oriented point cloud registration Summer 2023
- Zaki Cole (Undergrad): Automated data collection in CARLA simulator Fall 2022

## TECHNICAL SKILLS

**Languages:** C++, Python, MATLAB, C, C#, Java

**Systems & Tools:** Linux, Git, Docker, ROS, CARLA, AirSim, Unity, PyTorch, TensorFlow